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Mineralogic assessment of natural radionuclide rich beach sands in Western Anatolia (Canakkale)

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Abstract

Natural radioactivity concentrations in beach deposits and surrounding magmatic rocks at Ezine, Çanakkale, are notably high. The specific activities of ^{238}U , ^{232}Th and ^{40}K are in the range of 40-361, 50-341 and 670-1572 Bq kg⁻¹, respectively, for the magmatic rock suites. Bulk sand samples from the beaches exhibit even higher concentrations of natural radioactivity: Hantepe beach, located near the contact of Kestanbol pluton with pre-Tertiary basement units displays extremely high ^{238}U (1885 Bq kg⁻¹) and ^{232}Th (4360 Bq kg⁻¹) with the lowest ^{40}K (687 Bq kg⁻¹) activity concentrations.

This enrichment at Hantepe beach was attributed to the U-bearing accessory minerals (zircon, apatite, thorite, uranothorite, monazite, etc.) selectively rich along the northern margin of Kestanbol Pluton. These minerals serve as Rare Earth Elements (REE) repositories, prompting an extensive investigation into the economic potential of radionuclide-rich beach sands at Ezine.

In this study, gamma ray spectrometry measurements were conducted on fractions of beach sands to identify the source minerals of the radioactivity. The sand fraction concentrated in diamagnetic and felsic minerals (K-feldspar + Plagioclase ± Quartz) with grain sizes $\geq 250 \mu$, exhibits the highest ^{40}K specific activity (1173.3 ± 133.5 Bq kg⁻¹) but the lowest concentrations of ^{238}U (176.4 ± 18.1 Bq kg⁻¹) and ^{232}Th series

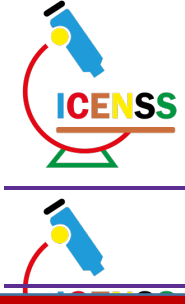
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($197.8 \pm 21.9 \text{ Bq kg}^{-1}$). On the contrary, the fraction rich in paramagnetic and mafic minerals (Pyroxene + Amphibole + Biotite + Epidote $\pm$ Granat) with the grain sizes $< 125 \mu\text{m}$ has the most elevated concentrations of ^{238}U ($884.5 \pm 56.6 \text{ Bq kg}^{-1}$) and ^{232}Th ($1332.9 \pm 99.1 \text{ Bq kg}^{-1}$) series and slightly elevated ^{40}K ($436.4 \pm 50.3 \text{ Bq kg}^{-1}$), compared to the Earth's average. These findings suggest that the rock-forming minerals in the magmatic rocks are also responsible for the natural high radioactivity in the region.

Keywords: Natural radioactivity, Beach Sands, Magmatic Rocks, Rare Earth Elements, Gamma-Spectroscopy





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Batı Anadolu'da (Çanakkale) radyonüklid bakımından zengin doğal plaj kumlarının mineralojik değerlendirilmesi

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Öz

Batı Anadolu'da Ezine (Çanakkale) dolaylarındaki plajlara ait kumların ve plajların yakınındaki magmatik kayaların doğal radyoaktivite konsantrasyonları oldukça yükectir. Magmatik kaya grupları için ^{238}U serisi, ^{232}Th serisi ve ^{40}K spesifik aktiviteleri sırasıyla 40-361, 50-341 ve 670-1572 Bq kg^{-1} aralığındadır. Plajlardan derlenen dökme kum örneklerinde ise radyoaktivite konsantrasyonların daha da yüksek olduğu ortaya konulmuştur: Kestanbol plütunu ile Tersiyer öncesi temel kayaların sınırına yakın bir alanda konumlanmış olan Hantepe plajına ait kum örnekleri en yüksek ^{238}U serisi (1885 Bq kg^{-1}) ve ^{232}Th serisi (4360 Bq kg^{-1}) ile en düşük ^{40}K (687 Bq kg^{-1}) aktivite konsantrasyonlarına sahiptir.

Hantepe sahilindeki bu zenginleşme, Kestanbol Plütunu'nun kuzey kenarı boyunca seçici olarak zengin olduğu öne sürülen Uranyumlu aksesuar minerallere (zirkon, apatit, torit, uranotorit, monazit, vb.) atfedilmiştir. Bu mineraller, yüksek Nadir Toprak Elementleri (NTE) içeriklerine sahip olduklarından, Ezine'deki radyonüklid zengin plaj kumlarının ekonomik potansiyelleri önem kazanmıştır.

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Bu çalışmada, radyoaktivitenin kaynak minerallerini belirlemek için pırlajlardan derlenen dökme kum örneklerinden elde edilen fraksiyonlar üzerinde gama ışıını spektrometresi ölçümleri yapılmıştır. Tane boyutları $\geq 250 \mu\text{m}$ olan, diyamanyetik ve felsik minerallerin (K-feldispat+ Plajiyoklaz $\pm$ Kuvars) yoğun olduğu fraksiyonda ^{40}K spesifik aktivitesinin ($1173.3 \pm 133.5 \text{ Bq kg}^{-1}$) en yüksek değerlere ulaştığı, ^{238}U ($176.4 \pm 18.1 \text{ Bq kg}^{-1}$) ve ^{232}Th serisinin ($197.8 \pm 21.9 \text{ Bq kg}^{-1}$) ise en düşük değerlerde olduğu belirlenmiştir. Bununla birlikte, tane boyu $< 125 \mu\text{m}$ olan, paramanyetik ve mafik minerallerce (Piroksen + Amfibol + Biyotit + Epidot $\pm$ Granat) zengin olan fazda ise ^{238}U ($884.5 \pm 56.6 \text{ Bq kg}^{-1}$) ve ^{232}Th ($1332.9 \pm 99.1 \text{ Bq kg}^{-1}$) serisi spesifik aktivitesinin en yüksek değerlere ulaştığı, ^{40}K spesifik aktivitesinin ise ($436.4 \pm 50.3 \text{ Bq kg}^{-1}$), Dünya ortalamasının hafifçe üzerinde olduğu tespit edilmiştir. Bu bulgular, magmatik kayalardan türeyerek plaj kumlarına katılan kayaç yapıcı minerallerin de bölgedeki yüksek doğal radyoaktivite değerlerinden sorumlu olduğunu gösterir.

Keywords: Doğal radyoaktivite, Plaj Kumu, Magmatik Kaya, Nadir Toprak Elementler, Gama Spektrometresi

Suppl 2. Calcium Data

		Pre-flight		In-flight		R+0		R+1		R+2-7		>R+7	
#	Author(s), year (units)												
(n)	Condition	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$
		d	Trend	d	Trend	d	Trend	d	Trend	d	Trend	d	Trend

Trend legend:

(↓) authors found significant decrease from pre-flight ($p < 0.05$)

(▼) authors found trend to significant decrease from pre-flight ($0.05 < p \leq 0.1$)

(↑) authors found significant increase from pre-flight ($p < 0.05$)

(▲) authors found trend to significant increase from pre-flight ($0.05 < p \leq 0.1$)

(-) authors found nonsignificant change from pre-flight

(U) p-values unreported by authors.

Table S1. Data table with mean serum calcium data, standard deviation, sample size, experimental condition, and significance of results extracted from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Smith et al. 2005 (mmol/L)												
	(6) Real	2.4	0.1	2.3	0.2	2.4	0.1	2.4	0.1	2.3	0.1	2.3	0.1
		1	7	2	0	4	2	4	2	5	7	6	2
				-	-	0.0	-	0.0	-	-	-	-	-
				0.4		7		7		0.2		0.2	
				3						9		7	
	(16) Real	2.4	0.2			2.4	0.3	2.4	0.0	2.3	0.4	2.3	0.2
		1	4			4	6	4	4	7	4	6	4
						0.1	-	0.1	-	0.1	-	-	-
						0		7		4		0.2	
												4	
2	Shackelford et al. 2004 (mg/dL)												
	(18) Simulated	9.6	1.0	9.7	0.8								
			6		9								
				0.1	-								
				0									
3	Linossier et al. ⁵ 2022 (mg/L)	96.0	4.7	99.6	6.8					96.4	7.1		
	(9) Simulated			0.6	↑					0.0	▲		
				4						7			
5	Smith et al. 2015 (mmol/L)												
	(7) Real	2.3	0.1	2.2	0.1								
		4		8	4								
				-	-								
				0.4									
				9									
6	Armbrecht et al. 2010 (per cent change from baseline)												

⁺Results are reported as medians with IQRs. Standard deviations are approximated from IQRs using the formula $SD = IQR/1.35$ as given by the Cochrane Handbook for Systematic Reviews.

	(10) Simulated	0	N/A	- 0.8 9	0.9 4			-3.4	1.3	-1.7	1.0	-0.9	1.1
				- 0.4 3	--			- 2.9 3	↓	- 1.7 0	--	- 0.8 6	--
7	Smith et al. 2012 (mmol/L)												
	(6) Real	2.3	0.1	2.3	0.1	2.3	0.2					2.3	0.1
				0.0 0	--	0.0 0	--					0.0 0	--
8	Zerwekh et al. 2007 (mmol/L)												
	(10) Simulated	2.3 5	0.0 9	2.3 8	0.1 0								
				0.3 2	--								
9	Smith et al. 2003 (mmol/L)												
	(8) Simulated	2.3 4	0.0 7	2.3 8	0.0 8			2.3 6	0.0 7	2.3 5	0.1 2		
				0.5 3	↑			0.2 9	--	0.1 0	--		
10	Zwart et al. 2007 (mmol/L)												
	(7) Simulated	2.3 4	0.1 0	2.3 7	0.1 4	2.4 0	0.1 3	2.3 4	0.1 4	2.3 2	0.1 5		
				0.2 5	--	0.5 2	--	0.0 0	--	- 0.1 6	↓		
11	Morgan et al. 2012 (mmol/L)												
	(12) Simulated	2.3 2	0.0 7	2.3 8	0.0 8					2.2 9	0.0 7		
				0.8 0	↑					- 0.4 3	--		
12	Rittweger et al. 2005 (mmol/L)												
	(25) Simulated	2.5 8	0.0 6	2.6 2	0.0 4			2.4 7	0.0 3	2.4 8	0.0 2	2.5 4	0.0 4
				0.9 3	--			- 2.3 2	↓	- 2.2 4	↓	- 1.3 7	--

Table S2. Data table with mean urinary calcium data, standard deviation, sample size, experimental condition, and significance of results extracted from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Smith et al. 2005 (mmol/day)												
	(6) Real	5.3	1.4	7.9	3.8	2.7	1.7	6.5	1.4	4.7	1.7	5.3	1.0
				0.91	▲	-	-	0.86	-	-	-	0.00	-
	(16) Real	4.8	1.7			2.7	1.7	5.4	1.6	5.4	1.9	4.2	1.5
						-	↓	0.36	-	0.36	-	-	-
						1.24						0.37	
2	Smith et al. 2015 (mmol/day)												
	(7) Real	4.5	2.5	6.8	3.9								
				0.70	↑								
3	Zerwekh et al. 2007 (mg/kg/day)												
	(10) Simulated	116	65	180	87								
				0.83	↑								
4	Smith et al. 2003 (mmol/day)												
	(8) Simulated	5.67	3.19	7.48	3.35			5.66	2.76				
				0.55	↑			0.00	-				
5	Zwart et al. 2007 (mmol/day)												
	(7) Simulated	3.95	1.32	6.64	2.54			5.22	2.83				
				1.33	↑			0.58	-				
6	Morgan et al. 2012 (mmol/day)												
	(12) Simulated	5.5	2.1	7.0	2.5			6.3	2.0	5.7	1.8		
				0.65	↑			0.39	↑	0.10	-		

Table S3. Data table with mean ionized calcium data, standard deviation, sample size, experimental condition, and significance of results from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Shackelford et al. 2004 (mmol/L)												

	(18) Simulated	1.2	0.01	1.3	0.01								
				1.0	↑								
2	Smith et al. 2012 (mmol/L)												
	(6) Real	1.24	0.04			1.17	0.06					1.21	0.04
						- 1.37	↓					- 0.75	-
3	Morgan et al. 2012 (mmol/L)												
	(12) Simulated	1.23	0.04	1.23	0.04					1.23	0.03		
				0.00	-					0.00	-		

